Spring Boot Step-by-Step Execution Guide

From JDBC to Spring Boot with Storytelling + Hands-on

Step 1: Install Required Software

- Install Java JDK 17 or above
- Install IntelliJ IDEA or VS Code
- Install MySQL Database
- Install Postman for API testing
- Install Maven (optional if IDE handles it)
- Official Downloads:
- • Spring Initializr: https://start.spring.io
- • IntelliJ IDEA: https://www.jetbrains.com/idea/

Step 2: Understanding the Story

- Core Java = Small Local Shop
- JDBC = Delivery Clerk handling database communication
- Spring = Intelligent City Management System
- Spring Boot = Fully Automated Smart City

Step 3: Create Your First Spring Boot Project

- Go to Spring Initializr
- Choose Maven Project
- Select Java Language
- Add Dependencies: Spring Web, Spring Data JPA, MySQL Driver
- Generate and Download the project zip
- Open it in IntelliJ or VS Code

Step 4: Understanding Project Structure

- src/main/java = Java code
- controller package = handles requests
- service package = business logic
- repository package = database layer
- resources/application.properties = configuration file

Step 5: Create Your First API

- Create a Controller class
- Use @RestController annotation
- Use @GetMapping to create endpoints
- Run the application and open browser

Step 6: What is Spring JPA?

- JPA = Java Persistence API
- It helps Java objects connect with database tables
- No need to write repetitive SQL manually
- Spring Data JPA simplifies database operations

Step 7: Creating Entity Class

- @Entity tells Spring this class maps to a table
- @Id marks the primary key
- Spring automatically creates table structure

Step 8: Repository Layer

- Repository handles database communication
- Create interface extending JpaRepository
- Spring automatically provides CRUD methods
- save(), findAll(), deleteById() etc.

Step 9: Service Layer

- Service layer contains business logic
- Controller talks to Service
- Service talks to Repository
- This creates clean architecture

Step 10: Connect Database

- Open application.properties
- Add database URL, username, password
- Spring Boot automatically connects to MySQL

Step 11: Run the Application

- Click Run button in IntelliJ/VS Code
- Tomcat server starts automatically
- Open browser or Postman
- Test APIs using localhost:8080

Step 12: Real Industry Connection

- Companies use Spring Boot for enterprise applications
- Used in banking, e-commerce, healthcare, logistics
- Skills learned: APIs, security, databases, deployment